

Knowledge Representation & Reasoning in AI

Part 1	Part 2	Part 3	Part 4	Part 5
Introduction to KR	Types of Knowledge	Propositional Logic	First Order Logic	Wumpus World

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What Is Artificial Intelligence?

Before we study KRR, we must understand what AI is — and what it is trying to achieve.

Definition

Artificial Intelligence (AI) is the field of computer science concerned with building systems that can perform tasks which, if performed by a human, would require intelligence. These tasks include perceiving the environment, learning from experience, reasoning about knowledge, solving problems, and taking actions to achieve goals.

PERCEIVE

Sense and interpret data from the environment — images, text, speech, sensor readings.

LEARN

Identify patterns in data and improve performance over time without being explicitly programmed.

REASON

Use stored knowledge and logic to draw conclusions, solve problems, and make decisions.

ACT

Take intelligent actions in the world — recommendations, decisions, movements, responses.

The Major Branches of AI

AI is not one thing — it is a family of sub-fields, each tackling a different aspect of intelligence.



Machine Learning

Systems that learn from data — finding patterns without explicit programming. Includes supervised, unsupervised, and reinforcement learning.



Natural Language Processing

Understanding and generating human language — chatbots, translation, sentiment analysis, and large language models like GPT.



Robotics & Planning

AI agents that interact physically with the world — navigation, manipulation, task planning in dynamic environments.



Expert Systems

Rule-based AI that encodes domain expertise (medical, legal, financial) as IF-THEN rules to give professional-level advice.



Deep Learning

Neural networks with many layers that learn hierarchical representations of data. Powers image recognition, speech, and language models.



Computer Vision

Enabling machines to interpret visual information — object detection, facial recognition, medical imaging, self-driving cars.



Knowledge Representation & Reasoning

THIS

Storing facts and rules formally, then applying logic to derive conclusions. THIS is what this course is about — the brain's reasoning engine.



Planning & Search

Finding sequences of actions to achieve goals in complex state spaces — used in game AI, logistics, and autonomous systems.

The World of AI — Where Knowledge Representation & Reasoning Lives

ARTIFICIAL INTELLIGENCE

Computer Vision 📷

Speech Recognition 🎤

Sensor Fusion 📡

NLP (Input) 💬

ACTION

Robotics 🤖

Planning & Search 🗺️

Decision Making ⚡

Natural Language Gen. 📝

★ KNOWLEDGE REPRESENTATION & REASONING

Knowledge Base (KB)

Facts + Rules stored formally

Propositional Logic

T/F propositions, connectives

First Order Logic (FOL)

$\forall \exists$ Variables, Predicates

Inference Engine

Modus Ponens, Resolution

Ontologies & Semantic Web

OWL, RDF, Knowledge Graphs

Expert Systems

Medical, Legal, Financial AI

LEARNING

Machine Learning 📊

Deep Learning 🔗

Reinforcement Learning 🎮

Neural Networks 🧬

APPLICATIONS

Medical AI 🏥

Autonomous Vehicles 🚗

Search Engines 🔍

Expert Systems 🧠

What Is This Course About?

The central question: How does a machine know things — and how does it think?

Knowledge Representation and Reasoning (KRR) answers one of the most fundamental questions in Artificial Intelligence: how does a machine know things — and how does it think?

THE CENTRAL PROBLEM

Imagine building an AI doctor. You have millions of patient records.

A pile of records is NOT intelligence. The AI needs to:

① **REPRESENT** what it knows (store facts & rules in a structured form) ② **REASON** over that knowledge (derive new conclusions) ③ **ACT** intelligently (make decisions, give recommendations)

Raw data → Structured knowledge → Intelligent reasoning

WHY #1: EXPLAINABILITY

Neural networks are black boxes. Regulators in healthcare, finance & law require explainable AI — KRR provides transparent, auditable reasoning chains.

WHY #2: RELIABILITY

Aviation, medical devices, autonomous vehicles — systems that must be formally verified. You can PROVE an AI will never recommend a drug a patient is allergic to.

WHY #3: NEURO-SYMBOLIC AI

The cutting edge combines neural networks (perception & learning) with symbolic reasoning (logic & planning). DeepMind & IBM are actively hiring for this.

WHY #4: FOUNDATION

Prolog, ontologies, semantic web, planning systems, expert systems, knowledge graphs — all direct applications of this course. Without this, you are a user, never a designer.

How KRR Powers Real AI Systems You Already Know

Google Search / Knowledge Graph

Declarative KR — billions of facts stored as entity-relationship triples

Technology

Siri / Alexa / Google Assistant

FOL-style intent parsing + rule-based inference to trigger actions

Consumer AI

Medical Diagnosis AI (IBM Watson)

Expert systems — IF-THEN rules derived from clinical knowledge bases

Healthcare

Email Spam Filters

Heuristic rules + propositional logic to classify T/F properties

Cybersecurity

Chess & Game AI (Stockfish)

Heuristic knowledge + forward chaining over symbolic game states

Gaming

Self-Driving Cars

Sensor percepts → symbolic world model → FOL reasoning → safe action

Automotive

Prolog & Logic Programming

Direct implementation of FOL with backward chaining inference

Software / AI
Dev

Legal & Compliance AI

Production rule systems encoding laws as IF-THEN logical statements

Finance / Legal

Semantic Web (OWL, RDF)

FOL-based ontologies for interoperable AI knowledge across systems

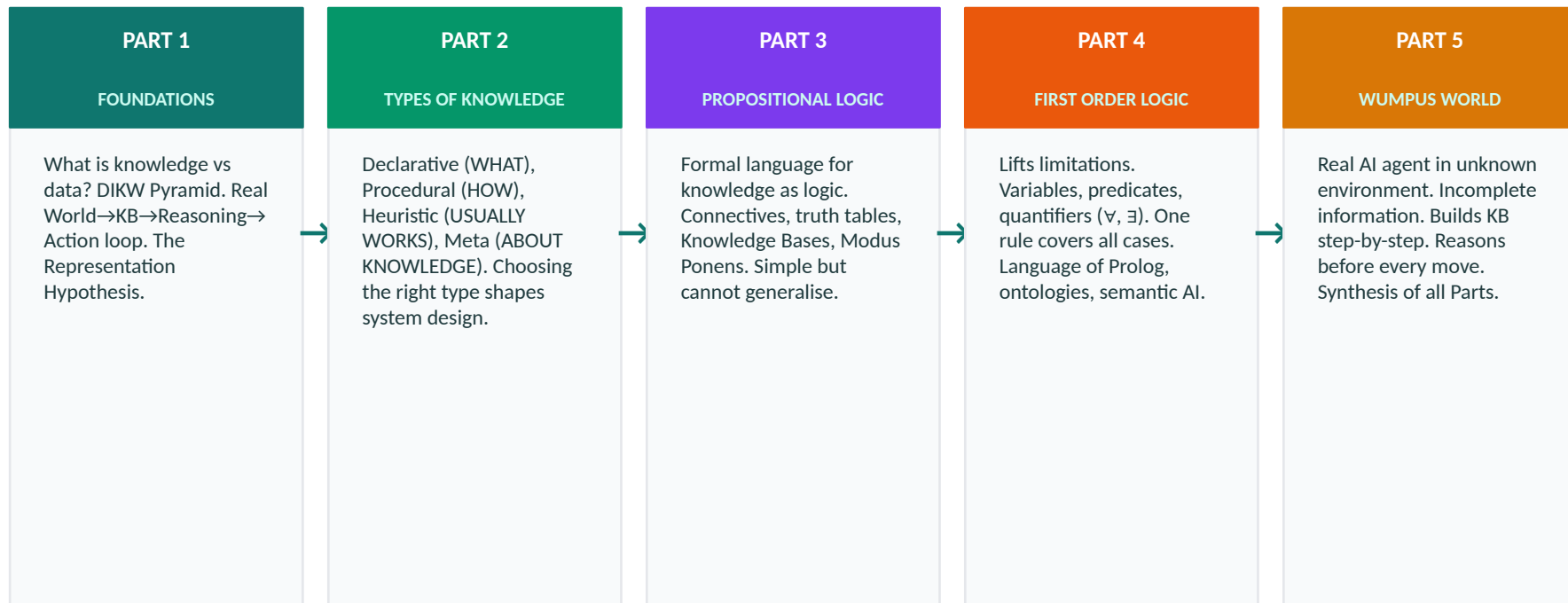
Web / Data
Science

Fraud Detection Systems

Meta-knowledge + heuristic rules for confidence-scored risk assessment

Banking /
FinTech

The Learning Journey — How Every Topic Connects



OUTCOME: You can design, build, query, and reason over a real Knowledge Base. You understand not just HOW but WHY each technique works.

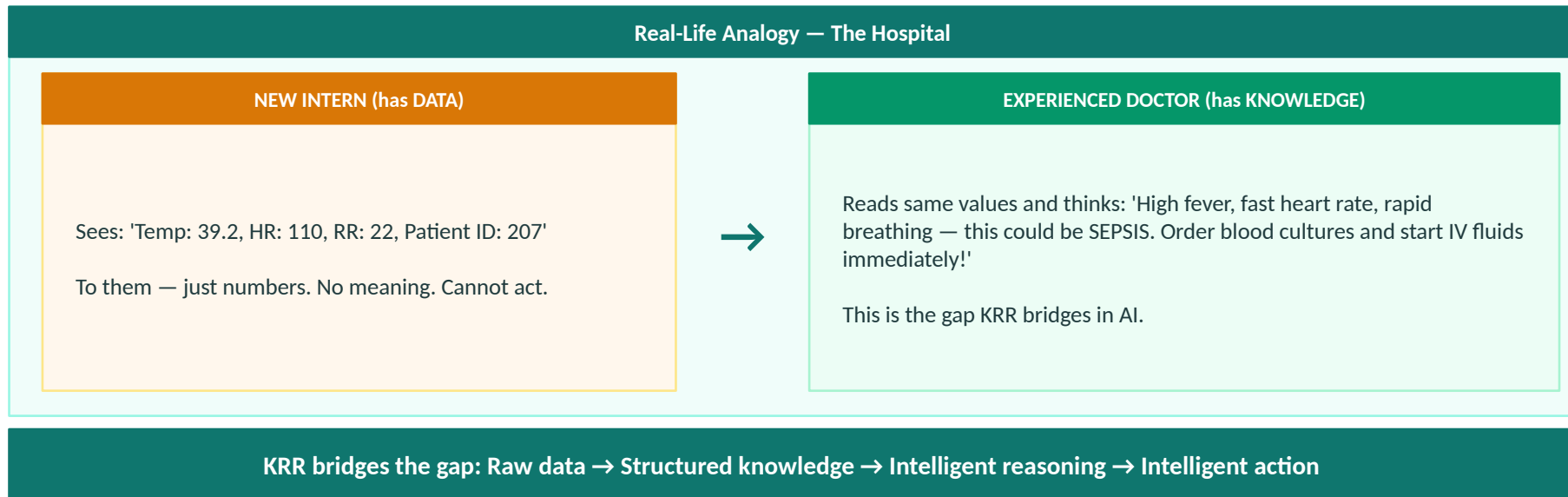
01

Introduction to Knowledge Representation

1.1 What Is Knowledge in AI?

Knowledge is information that has been understood, organised, and can be used to solve problems or make decisions.

Simple answer: Knowledge is information that can be USED. Raw data cannot be acted on directly. Knowledge can.



Critical Thinking Questions: When is symbolic KRR the right approach? What happens when the KB is incomplete? How does expressive power trade off against computational cost?

1.2 The DIKW Pyramid

A foundational model showing how raw facts evolve into wisdom — layer by layer.

D — DATA

What: Raw, unprocessed numbers or symbols with no meaning on their own.

Example: Temp=39.2, HR=110, RR=22, Patient=207

I — INFORMATION

What: Data given context and meaning.

Example: Patient 207 has a high fever, fast heart rate, and rapid breathing.

K — KNOWLEDGE

What: Information combined with rules and patterns.

Example: Fever+HR>100+RR>20 → possible sepsis. Start blood cultures NOW.

W — WISDOM

What: Applying knowledge with judgment and experience.

Example: Start antibiotics before cultures return — every hour delay raises mortality 7%.

Class Activity 1 — DIKW Classification Game

Work in pairs — Smart Traffic System

Classify each statement as: Data / Information / Knowledge / Wisdom

1. Vehicle count at Signal A: 47

2. Signal A is congested right now.

3. If a signal has more than 40 vehicles, extend green light by 30 seconds.

4. Route via Main Boulevard will save 12 minutes compared to Canal Road.

5. GPS coordinates: 31.5204, 74.3587

6. Speed camera recorded 94 km/h in a 60 km/h zone.

7. Extending green on Ferozpur Road before Friday prayer reduces gridlock by 35%.

8. Car with plate LHR-441 is travelling at 94 km/h.

Class Activity 1 — Solution

DIKW Classification — Smart Traffic System

Answers:

1. Vehicle count: 47

DATA

2. Signal A is congested

INFORMATION

3. IF vehicles > 40 THEN extend green 30 sec

KNOWLEDGE

4. Route via Main Blvd saves 12 min

INFORMATION

5. GPS: 31.5204, 74.3587

DATA

6. Car going 94 km/h in 60 zone

INFORMATION

7. Extending green before Friday prayer → 35% less gridlock

WISDOM

8. LHR-441 is speeding

INFORMATION

1.3 Real World → Knowledge → Reasoning → Action

The fundamental loop every AI reasoning system follows.



Example 1: Medical Diagnosis AI

Real World: Patient: temperature 39.8, cough, fatigue for 3 days

KB Rule 1: IF fever>38 AND cough AND fatigue THEN likely_flu

KB Rule 2: IF likely_flu THEN recommend_antiviral

Reasoning: fever=39.8>38 ✓, cough=TRUE ✓, fatigue=TRUE ✓

Apply Rule 1 ⇒ likely_flu=TRUE [Modus Ponens]

Apply Rule 2 ⇒ recommend_antiviral=TRUE

Action: 'Likely diagnosis: Influenza. Prescribe Tamiflu + rest.'

Example 2: Spam Filter AI

Real World: Email: subject='YOU WON \$1,000,000!!!',
sender=unknown@xyz

KB Rule 1: IF subject_has_large_sum AND sender_unknown THEN
spam=TRUE

KB Rule 2: IF ALL_CAPS_word_count > 3 THEN spam_score += 2

Reasoning: subject_has_large_sum=TRUE, sender_unknown=TRUE
⇒ spam=TRUE

Action: Move email to Spam folder automatically.

1.4 The Representation Hypothesis

One of the most important theoretical ideas in the history of AI — Newell & Simon, 1976.

Physical Symbol System Hypothesis (Newell & Simon, 1976)

"A physical symbol system has the necessary and sufficient means for general intelligent action. Any system capable of general intelligent behaviour is a physical symbol system."

Translation: Intelligence = symbols + rules. You do not need to simulate every atom of reality. You just need to correctly represent the relevant parts as symbols and define rules that operate on them.

SYMBOLS

Internal representations of objects, relations, states, rules
Examples: Ali, CS701, enrolled(x,y), Human(x)

+

SYMBOL MANIPULATION

Operations on symbols according to formal rules
Examples: Modus Ponens, Unification, Inference

+

INTELLIGENCE

Emerges from well-chosen symbols + correct rules — no 'magic' needed
This is the foundation of ALL rule-based AI systems

Practice Questions — Part 1

Think carefully and write your answers before turning to the next slide.

Q1. What is the difference between data, information, and knowledge? Use a real-world traffic example to illustrate each level.

Q2. Classify each of the following as Data, Information, Knowledge, or Wisdom:

(a) Marks = 85 (b) Student failed 3 subjects (c) IF failed > 2 THEN put on probation (d) Mentoring raises pass rate by 78%

Q3. Explain the Representation Hypothesis (Newell & Simon, 1976) in your own words. Why does it matter for building AI systems?

Q4. Describe the full Real World → Knowledge → Reasoning → Action loop for a university admission system. Include sample facts, rules, and an inference step.

Practice Questions — Part 1: Answers

Q1. Data vs Information vs Knowledge (traffic example)

Data: raw numbers (speed=94 km/h). Information: car LHR-441 is speeding in a 60 zone. Knowledge: IF speed>60 THEN speeding=TRUE; issue fine. The progression adds context, meaning, and actionable rules.

Q2. Classify: (a) Marks=85 (b) Student failed 3 subjects (c) IF failed>2 THEN probation (d) Mentoring raises pass rate 78%

(a) DATA — raw number. (b) INFORMATION — data + context applied. (c) KNOWLEDGE — rule derived from patterns. (d) WISDOM — experience + judgment + context.

Q3. Representation Hypothesis

The Hypothesis says intelligent behaviour can emerge from manipulating symbolic structures using formal rules — no physical simulation needed. It justifies building AI using symbols (facts) + rules (logic). Without it, we would need to simulate every atom of reality inside a computer.

Q4. Real World → Action loop for university admission

Real World: Student applies: CGPA=3.7, Test=85, Activities=5. Knowledge: IF CGPA $\geq$ 3.5 AND Test $\geq$ 80 AND Activities $\geq$ 3 THEN admit=TRUE. Reasoning: 3.7 $\geq$ 3.5 ✓, 85 $\geq$ 80 ✓, 5 $\geq$ 3 ✓ $\Rightarrow$ admit=TRUE [Modus Ponens]. Action: System sends admission offer letter automatically.

02

Types of Knowledge

Declarative (WHAT) • Procedural (HOW) • Heuristic (USUALLY WORKS) •
Meta (ABOUT KNOWLEDGE)

Choosing the right type is critical for effective system design.

The Four Types of Knowledge — Overview

DECLARATIVE

Answers: *WHAT* is true?

Facts about the world — stored as assertions that are simply true or false. The foundation of any knowledge base.

Examples:

- Paris is the capital of France.
- Water boils at 100°C at sea level.
- `enrolled(Ali, CS701)`
- `credit_hours(CS701, 3)`

PROCEDURAL

Answers: *HOW* to do it?

Step-by-step processes to accomplish tasks. Describes sequences of operations.

Examples:

- Step 1: Subtract 6 from both sides: $3x=15$
- Step 2: Divide by 3: $x=5$
- ATM: Insert → PIN → Amount → Dispense
- Triage: Check breathing → If not, start CPR

HEURISTIC

Answers: *What USUALLY* works?

Experience-based practical guidance. Can be wrong — that is what makes it heuristic. Rule of thumb, not guarantee.

Examples:

- Patient 50+, smoker, chest pain → check heart first
- Spam: 3+ exclamation marks + dollar amount → likely spam
- Chess: control centre in opening phase
- $RSI > 70$ + volume spike → possibly overbought

META-KNOWLEDGE

Answers: *What do I KNOW* about my knowledge?

Knowledge about the scope, reliability, or source of other knowledge. Enables self-aware AI systems.

Examples:

- 'My flu diagnosis rule is 87% accurate on adults.'
- 'I do not have enough data for this rare disease.'
- 'Lab result from 3 days ago may be outdated.'
- Confidence scoring: 'Diagnosis: Flu (87%)'

Class Activity 3 — Knowledge Sorting: Ride-Sharing App

Classify each item as: Declarative / Procedural / Heuristic / Meta

1. Distance from Model Town to Airport is 22 km.

2. To calculate fare: $\text{base_fare} + (\text{distance} \times \text{rate_per_km}) + \text{surge_multiplier}$

3. If it is raining, surge pricing usually increases by 1.5×.

4. Distance estimation model has error margin of ± 0.8 km.

5. Driver Ali has a 4.8 star rating.

6. If a driver cancels more than 3 trips in an hour, flag their account.

7. Friday evenings see 3× more ride requests near food streets.

8. Our surge-pricing prediction is 74% accurate for weekends.

9. To assign a driver: find nearest available within 5 km radius.

10. GPS module data is 15 minutes old — refresh before routing.

Class Activity 3 — Solution

Ride-Sharing App — Knowledge Type Classification

1. Distance = 22 km

DECLARATIVE

2. Fare formula

PROCEDURAL

3. Rain → surge usually 1.5×

HEURISTIC

4. Error margin ± 0.8 km

META

5. Driver Ali rating: 4.8

DECLARATIVE

6. IF cancels > 3/hr THEN flag

PROCEDURAL

7. Friday evenings = 3× demand near food streets

HEURISTIC

8. Surge prediction 74% accurate

META

9. Algorithm to assign nearest driver

PROCEDURAL

10. GPS data is 15 min old — refresh

META

Practice Questions — Part 2

Think carefully and write your answers before turning to the next slide.

Q5. What is the key difference between declarative and procedural knowledge? Can declarative knowledge be converted into procedural knowledge? Give an example.

Q6. Give 3 examples of heuristic knowledge from your own experience as a student or developer. For each, explain why it is heuristic and NOT a guaranteed rule.

Q7. You are building an AI chatbot for a bank. Classify each of the following items as Declarative, Procedural, Heuristic, or Meta-Knowledge:
(a) Customer balance: Rs 45,000 (b) Transfer process: verify PIN → check balance → deduct → credit (c) Transfers of Rs 50,000+ at midnight are often fraudulent (d) Our fraud detection model has 93% precision (e) Minimum balance rule: Rs 1,000 (f) 3 failed PINs in a row → lock account

Practice Questions — Part 2: Answers

Q5. Declarative vs Procedural knowledge

Declarative: WHAT is true. e.g., 'A sorted array has each element $\leq$ the next.'

Procedural: HOW to do it. e.g., 'Compare neighbours, swap if out of order, repeat (bubble sort).'

Conversion: Yes — the declarative goal 'sorted array' becomes the procedure. AI systems often need both: declare the GOAL, then procedurally reach it.

Q6. Three heuristic knowledge examples (student/developer)

1. 'If a question has a very long answer in the key — skip it in the exam, do short ones first.' (Not guaranteed — long questions may be easier for you.)
2. 'If the professor repeats a concept 3+ times — it will be in the exam.' (Not guaranteed — may just be addressing student confusion.)
3. 'If code ran fine yesterday and broke today — I probably changed something I forgot.' (Not guaranteed — could be an external library update or environment issue.)

Q7. Bank chatbot classification

(a) DECLARATIVE — stored fact. (b) PROCEDURAL — deterministic step-by-step process. (c) HEURISTIC — experiential pattern (can be wrong). (d) META — reliability/accuracy of our own system. (e) DECLARATIVE — business policy fact. (f) PROCEDURAL — exact deterministic rule.

03

Propositional Logic

The foundation of formal reasoning in AI. A precise, mathematical language to express knowledge and draw conclusions.

Propositions • Connectives • Truth Tables • Knowledge Bases • Modus Ponens

3.1 Propositions & Connectives (AND, OR, NOT)

A proposition is any statement definitively TRUE or FALSE — not a question, command, or vague statement.

✓ VALID PROPOSITIONS (True or False)

- It is raining outside.
- Karachi is the capital of Pakistan. (FALSE)
- $2 + 2 = 4$. (TRUE)
- Ali passed the exam.
- Python is a programming language. (TRUE)

✗ NOT Valid Propositions

- Is it raining? (Question)
- Please close the window. (Command)
- $x + y = 10$ (Depends on x, y)
- She is tall. (Vague — tall vs what?)
- Study harder! (Imperative)

$P \wedge Q$ — AND (Conjunction)

Both must be true for the whole statement to be true.

P='I submitted assignment.'
Q='I attended lab.'
 $P \wedge Q \rightarrow$ TRUE only if BOTH done.

P	Q	Result
T	T	T
T	F	F
F	T	F
F	F	F

$P \vee Q$ — OR (Disjunction)

At least one must be true. Both can be true — OR is inclusive.

P='Passed midterm.' Q='Passed final.'
 $P \vee Q =$ TRUE if at least one exam passed.

P	Q	Result
T	T	T
T	F	T
F	T	T
F	F	F

$\neg P$ — NOT (Negation)

Reverses truth value. TRUE becomes FALSE. FALSE becomes TRUE.

P='Server is online.'
 $\neg P =$ 'Server is offline.'

P	$\neg P$
T	F
F	T

Connectives: IMPLIES ($\rightarrow$) and BICONDITIONAL ($\leftrightarrow$)

Two of the most important — and most misunderstood — connectives in logic.

$P \rightarrow Q$ IMPLIES (Implication)

'If P then Q'. FALSE only when P is TRUE but Q is FALSE — the premise holds but the conclusion fails.

Think of it as a PROMISE:

Case 1: Rains AND road wet $\rightarrow$ Promise kept!

TRUE

Case 2 ★: Rains AND road DRY $\rightarrow$ Promise BROKEN!

FALSE

Case 3: No rain AND road wet $\rightarrow$ Something else wet it. Promise untested.

TRUE

Case 4: No rain AND road dry $\rightarrow$ Fine. Promise untested.

TRUE

KEY TIP: $P \rightarrow Q$ is ONLY false when $P=\text{TRUE}$ and $Q=\text{FALSE}$. All other combinations are TRUE. Memorise: 'A true premise leading to a false conclusion breaks the implication.'

$P \leftrightarrow Q$ BICONDITIONAL

'P if and only if Q' — TRUE when both have the SAME truth value.

Example: P ='You study hard.' Q ='You pass exam.'
 $P \leftrightarrow Q$ = 'You pass IF AND ONLY IF you study.'

- Pass without studying? FALSE
- Study but still fail? FALSE
- Both study+pass: TRUE
- Both no-study+fail: TRUE

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

3.3 Knowledge Base, Inference & Modus Ponens

A Knowledge Base is a collection of logical sentences (facts + rules) the AI believes to be true.

Knowledge Base — University Admission System

Fact 1: $\text{MatricScore}(\text{Ali}) = 82$

Fact 2: $\text{FSCScore}(\text{Ali}) = 79$

Fact 3: $\text{TestScore}(\text{Ali}) = 88$

Fact 4: $\text{Paid_Fee}(\text{Ali}) = \text{TRUE}$

Rule 1: IF $\text{Matric} \geq 80$ AND $\text{FSC} \geq 70$ AND $\text{Test} \geq 85$ THEN Eligible

Rule 2: IF Eligible AND Paid_Fee THEN Admitted

Modus Ponens — The Core Inference Rule

Premise 1: $P \rightarrow Q$ (If P then Q)

Premise 2: P (P is TRUE)

Conclusion: Q (Therefore Q must be TRUE)

Example:

IF $\text{Eligible}(\text{Ali}) \rightarrow \text{CanApply}(\text{Ali})$

$\text{Eligible}(\text{Ali}) = \text{TRUE}$

 $\text{CanApply}(\text{Ali}) = \text{TRUE} \checkmark$

Class Activity 5 — Home Security Inference Chain

Rules: R1: $\text{MotionDetected} \rightarrow \text{AlertTriggered}$ R2: $\text{AlertTriggered} \text{ AND } \text{NightTime} \rightarrow \text{SendSMS}$ R3: $\text{AlertTriggered} \text{ AND } \text{NightTime} \rightarrow \text{TurnOnLights}$ R4: $\text{SendSMS} \text{ AND } \text{OwnerAway} \rightarrow \text{CallPolice}$ R5: $\text{DoorUnlocked} \rightarrow \text{MotionDetected}$

Facts: $\text{DoorUnlocked} = \text{TRUE}$, $\text{NightTime} = \text{TRUE}$, $\text{OwnerAway} = \text{TRUE}$

Task: Trace the full inference chain step by step. What is the final action? Which rule fires at each step?

Class Activity 5 — Solution

Home Security Inference Chain — Facts: DoorUnlocked=T, NightTime=T, OwnerAway=T

Facts	DoorUnlocked=TRUE, NightTime=TRUE, OwnerAway=TRUE
Step 1	DoorUnlocked=T + Rule 5 (DoorUnlocked $\rightarrow$ MotionDetected) $\Rightarrow$ MotionDetected=TRUE
Step 2	MotionDetected=T + Rule 1 (MotionDetected $\rightarrow$ AlertTriggered) $\Rightarrow$ AlertTriggered=TRUE
Step 3	AlertTriggered=T AND NightTime=T + Rule 2 $\Rightarrow$ SendSMS=TRUE
Step 4	AlertTriggered=T AND NightTime=T + Rule 3 $\Rightarrow$ TurnOnLights=TRUE
Step 5	SendSMS=T AND OwnerAway=T + Rule 4 $\Rightarrow$ CallPolice=TRUE

Final Actions: SendSMS • TurnOnLights • CallPolice | Every step = Modus Ponens: IF rule AND premise THEN conclusion

Class Activity 4 — Truth Table Construction

P ='I have my laptop' Q ='WiFi is working' R ='I can do my assignment'

Formula 1: $(P \text{ AND } Q) \rightarrow R$

Build the full truth table: $P \mid Q \mid R \mid P \text{ AND } Q \mid (P \text{ AND } Q) \rightarrow R$
For which combination(s) of P, Q, R is this formula FALSE?

Formula 2: $\text{NOT } P \text{ OR } (Q \text{ AND } R)$

Build the full truth table: $P \mid Q \mid R \mid \text{NOT } P \mid Q \text{ AND } R \mid \text{NOT } P \text{ OR } (Q \text{ AND } R)$
For which rows is this formula TRUE?

Formula 3: $(P \rightarrow Q) \text{ AND } (Q \rightarrow R) \rightarrow (P \rightarrow R)$

This is called Hypothetical Syllogism. Is it always TRUE? Build the truth table to verify. What does this formula mean in plain English?

Class Activity 4 — Solution

Truth Table Construction — P='Laptop' Q='WiFi' R='Can do assignment'

Formula 1: $(P \text{ AND } Q) \rightarrow R$

P	Q	R	$P \wedge Q$	$(P \wedge Q) \rightarrow R$
T	T	T	T	T
T	T	F	T	F ★
T	F	T	F	T
T	F	F	F	T
F	T	T	F	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T

Only FALSE when $P=T, Q=T, R=F$ (true premise, false conclusion)

Formula 3: Hypothetical Syllogism

$(P \rightarrow Q) \text{ AND } (Q \rightarrow R) \rightarrow (P \rightarrow R)$

Result: TAUTOLOGY — all 8 rows are TRUE.

Plain English: If $A \rightarrow B$ and $B \rightarrow C$, then $A \rightarrow C$ directly. The Chain Rule of logic.

Why it matters: This is the basis of multi-step inference chains. If $\text{Rain} \rightarrow \text{Flood}$ and $\text{Flood} \rightarrow \text{Evacuate}$, then $\text{Rain} \rightarrow \text{Evacuate}$ works directly.

⚠ Limitation of Propositional Logic — Why We Need First Order Logic Next

To say 'All students need a student ID', propositional logic requires: $\text{NeedsID_Ali}=\text{TRUE}, \text{NeedsID_Sara}=\text{TRUE}, \dots$ (one per student!)
We CANNOT write a single general rule 'for all students...' → This is exactly why First Order Logic exists.

Practice Questions — Part 3: Propositional Logic

Think carefully and write your answers before turning to the next slide.

Q8. Which of these are valid propositions? For each, state TRUE, FALSE, or INVALID and explain:

(a) The speed of light is 300,000 km/s (b) Can AI think? (c) Pakistan won the 1992 Cricket World Cup (d) Close the textbook. (e) This statement is false.

Q9. Build the full truth table for: $(P \text{ OR } Q) \text{ AND } (\text{NOT } P)$

For which combination(s) of P and Q is this formula TRUE?

Q10. Apply Modus Ponens step by step to derive all conclusions:

(1) $\text{Rain} \rightarrow \text{Flood}$ (2) $\text{Flood} \rightarrow \text{Evacuate}$ (3) $\text{Evacuate} \rightarrow \text{CallArmy}$ Given: $\text{Rain} = \text{TRUE}$

Q11. Create your own small Knowledge Base with 3 rules and 2 facts for a LIBRARY SYSTEM. Show one complete inference using Modus Ponens.

Q12. CHALLENGE: Is the formula $P \rightarrow (Q \rightarrow P)$ a tautology? Build the full truth table and justify your answer. What is this axiom called?

Practice Questions — Part 3: Answers

Q8. Valid propositions?

(a) Valid $\rightarrow$ TRUE (factual & verifiable). (b) INVALID — it's a question, no truth value. (c) Valid $\rightarrow$ TRUE (historical fact). (d) INVALID — command, not a proposition. (e) The Liar Paradox $\rightarrow$ leads to contradiction, not a valid proposition in classical logic.

Q9. Truth table for (P OR Q) AND (NOT P)

T|T: T AND F = F | T|F: T AND F = F | F|T: T AND T = T $\leftarrow$ ONLY TRUE ROW | F|F: F AND T = F

This expression is TRUE only when P=FALSE and Q=TRUE.

Q10. Modus Ponens chain: Rain $\rightarrow$ Flood $\rightarrow$ Evacuate $\rightarrow$ CallArmy, Rain=TRUE

Step 1: Rain=T + Rule (1) Rain $\rightarrow$ Flood $\Rightarrow$ Flood=T

Step 2: Flood=T + Rule (2) Flood $\rightarrow$ Evacuate $\Rightarrow$ Evacuate=T

Step 3: Evacuate=T + Rule (3) Evacuate $\rightarrow$ CallArmy $\Rightarrow$ CallArmy=T

All conclusions: Flood=TRUE, Evacuate=TRUE, CallArmy=TRUE

Q11. Library System KB + Modus Ponens inference

Fact 1: BookOverdue(Ali)=T Fact 2: FineUnpaid(Ali)=T

Rule 1: BookOverdue(x) $\rightarrow$ ChargeFine(x) Rule 2: FineUnpaid(x) AND BookOverdue(x) $\rightarrow$ BlockBorrowing(x) Rule 3: BlockBorrowing(x) $\rightarrow$ SendReminder(x)

Inference: BookOverdue(Ali)=T + R1 $\Rightarrow$ ChargeFine(Ali)=T; Both facts + R2 $\Rightarrow$ BlockBorrowing(Ali)=T + R3 $\Rightarrow$ SendReminder(Ali)=T

Q12. Is P $\rightarrow$ (Q $\rightarrow$ P) a tautology?

P=T, Q=T: Q $\rightarrow$ P=T, P $\rightarrow$ T=T | P=T, Q=F: Q $\rightarrow$ P=T, P $\rightarrow$ T=T | P=F, Q=T: Q $\rightarrow$ P=F, F $\rightarrow$ F=T | P=F, Q=F: Q $\rightarrow$ P=T, F $\rightarrow$ T=T

YES — all rows TRUE. This is the 'Weakening' (or Absorption) axiom: if P is true, then knowing Q doesn't change that P is true.

04

First Order Logic (FOL)

Also called Predicate Logic or First Order Predicate Calculus.

Variables • Constants • Predicates • Functions • Quantifiers $\forall$ $\exists$

The most widely used formal language for representing knowledge in AI.

4.1 & 4.2 — Why FOL? Building Blocks

FOL extends propositional logic with variables, predicates, and quantifiers.

The Problem Propositional Logic CANNOT Solve

Goal: 'Every student who passes all courses gets a degree.'

Propositional Logic: Ali_passed_CS101 AND Ali_passed_CS201 AND... $\rightarrow$ Ali_gets_degree THEN Sara_passed_CS101 AND... $\rightarrow$ Sara_gets_degree (One rule per student — impossible for thousands!)

FOL — ONE rule covers ALL students: $\forall x: \text{Student}(x) \text{ AND PassedAll}(x) \rightarrow \text{GetsDegree}(x)$

CONSTANTS

- Socrates — A specific Greek philosopher
- Lahore — A specific city
- CS301 — A specific course
- Pakistan — A specific country

VARIABLES

- x — Could refer to any person or object
- y — Could refer to any student
- course — Could refer to any course

PREDICATES

- Human(Socrates) — Unary (1 arg)
- Enrolled(Ali, CS301) — Binary (2 args)
- Scored(Ali, CS301, 88) — Ternary (3 args)
- Teaches(DrAhmed, CS301) — Binary

FUNCTIONS

- FatherOf(Ali) $\rightarrow$ returns Ali's father
- AgeOf(Sara) $\rightarrow$ returns Sara's age
- CapitalOf(Pakistan) $\rightarrow$ returns 'Islamabad'
- Note: Functions return objects, not T/F

4.3 Quantifiers — $\forall$ Universal and $\exists$ Existential

$\forall$ Universal Quantifier

For All / For Every

$\forall x \text{ Human}(x) \rightarrow \text{Mortal}(x)$

= 'For every x, if x is Human, then x is Mortal.' = 'All humans are mortal.'

More Examples:

$\forall x \text{ Student}(x) \rightarrow \text{HasStudentID}(x)$

[Every student has a student ID]

$\forall x \text{ Bird}(x) \wedge \neg \text{Penguin}(x) \rightarrow \text{CanFly}(x)$

[All non-penguin birds can fly]

$\forall x \forall y \text{ Married}(x,y) \rightarrow \text{Married}(y,x)$

[Marriage is symmetric]

$\forall x \text{ Employee}(x) \rightarrow \text{HasSalary}(x)$

[Every employee has a salary]

$\exists$ Existential Quantifier

There Exists / For Some

$\exists x \text{ Student}(x) \wedge \text{CGPA}(x) > 3.9$

= 'There exists at least one student whose CGPA is above 3.9.'

More Examples:

$\exists x \text{ Doctor}(x) \wedge \text{Female}(x)$

[There exists at least one female doctor]

$\exists x \text{ Course}(x) \wedge \text{Credits}(x) = 4$

[Some course has 4 credit hours]

$\exists x \text{ City}(x) \wedge \text{Capital}(x, \text{Pakistan})$

[There is a city that is Pakistan's capital]

$\exists x \text{ Student}(x) \wedge \text{CGPA}(x) = 4.0$

[At least one student has a perfect GPA]

4.4 Complete Reasoning — The Socrates Proof

Universal Instantiation + Modus Ponens — the foundation of all FOL reasoning.

Socrates Syllogism — Full FOL Proof

KNOWLEDGE BASE:

KB1: $\forall x \text{ Human}(x) \rightarrow \text{Mortal}(x)$ [All humans are mortal]

KB2: $\text{Human}(\text{Socrates})$ [Socrates is human]

PROVE: $\text{Mortal}(\text{Socrates})$

Step 1 — Universal Instantiation:

Take KB1: $\forall x \text{ Human}(x) \rightarrow \text{Mortal}(x)$

Substitute $x = \text{Socrates}$:

$\text{Human}(\text{Socrates}) \rightarrow \text{Mortal}(\text{Socrates})$

Step 2 — Modus Ponens:

$\text{Human}(\text{Socrates}) \rightarrow \text{Mortal}(\text{Socrates})$ [Step 1]

$\text{Human}(\text{Socrates})$ [KB2]

$\text{Mortal}(\text{Socrates}) \leftarrow \text{PROVED! } \checkmark$

Class Activity 6 — FOL Translation

Translate the following UNIVERSITY sentences into First Order Logic. Define your predicates before writing the formula.

1. All FAST students study computer science.
2. Some teachers teach more than one course.
3. Every course has at least one student enrolled.
4. No student can enroll without passing its prerequisite.
5. There exists a student who passed all 4-year courses.
6. If a student fails more than 3 courses, they are placed on probation.
7. All professors who publish research get a salary increment.
8. Every student on probation has a counselor assigned.

Class Activity 7 — Build a Hospital Patient Management KB (Group Project, 4 students)

Define 6+ predicates • Write 4+ facts (specific patients/doctors/diseases) • Write 4+ rules using $\forall$ or $\exists$ • Write 3 queries and answer them from your KB • Think about: Patients, Doctors, Diseases, Symptoms, Prescriptions, Allergies, Tests

Class Activity 6 — Solution

FOL Translation — University Domain

Predicates: $Student(x)$, $Teacher(t)$, $Course(c)$, $StudiesCS(x)$, $Teaches(t,c)$, $Enrolled(s,c)$, $Prerequisite(p,c)$, $Passed(s,c)$, $FailCount(s,n)$, $OnProbation(s)$, $Professor(p)$, $Publishes(p)$, $GetsIncrement(p)$, $Counselor(c,s)$

1	$\forall x: Student(x) \text{ AND } AtFAST(x) \rightarrow StudiesCS(x)$
2	$\exists t: Teacher(t) \text{ AND } \exists c1, c2: Teaches(t, c1) \text{ AND } Teaches(t, c2) \text{ AND } c1 \neq c2$
3	$\forall c: Course(c) \rightarrow \exists s: Student(s) \text{ AND } Enrolled(s, c)$
4	$\forall s, c: Enroll(s, c) \rightarrow \forall p: Prerequisite(p, c) \rightarrow Passed(s, p)$
6	$\forall s: Student(s) \text{ AND } FailCount(s) > 3 \rightarrow OnProbation(s)$
7	$\forall p: Professor(p) \text{ AND } Publishes(p) \rightarrow GetsIncrement(p)$
8	$\forall s: OnProbation(s) \rightarrow \exists c: Counselor(c, s)$
Neg.1	$\exists x: Student(x) \text{ AND } AtFAST(x) \text{ AND NOT } StudiesCS(x)$

Class Activity 7 — Solution

Hospital Patient Management KB — Sample Reference Answer

Predicates + Facts

Predicates:

Patient(p), Doctor(d), Disease(dis), HasSymptom(p,sym),
Diagnosed(p,dis), Prescribes(d,p,med), AllergicTo(p,med),
Specialist(d,dis)

Facts:

Patient(Ali) Doctor(DrSara) Disease(Flu) Disease(Malaria)
HasSymptom(Ali,Fever) HasSymptom(Ali,Cough)
Specialist(DrSara,Flu) AllergicTo(Ali,Penicillin)

Rules + Queries

Rules:

R1: $\forall p$: Patient(p) AND HasSymptom(p,Fever) AND HasSymptom(p,Cough)
→ SuspectedFlu(p)
R2: $\forall p,d,m$: Prescribes(d,p,m) AND AllergicTo(p,m) → DangerousAllergy(p)
R3: $\forall p$: SuspectedFlu(p) → NeedsBloodTest(p)
R4: $\forall p,dis,d$: Patient(p) AND Diagnosed(p,dis) AND Specialist(d,dis) →
ReferTo(p,d)

Queries:

Q1: Ali suspected flu? HasSymptom(Ali,Fever)=T AND
HasSymptom(Ali,Cough)=T $\Rightarrow$ SuspectedFlu(Ali)=T ✓
Q2: Ali needs blood test? SuspectedFlu(Ali)=T + R3 $\Rightarrow$
NeedsBloodTest(Ali)=T ✓
Q3: If Diagnosed(Ali,Flu), who to refer? + R4 $\Rightarrow$ ReferTo(Ali,DrSara) ✓

Practice Questions — Part 4: First Order Logic

Think carefully and write your answers before turning to the next slide.

Q13. Translate each sentence into FOL. Define your predicates first:

(a) All birds can fly. (b) Some birds cannot fly. (c) Penguins are birds that cannot fly. (d) Every animal that cannot fly still lives somewhere.

Q14. Given this FOL KB, answer each query with a full inference trace:

$\forall x \text{ Pakistani}(x) \rightarrow \text{SpeaksUrdu}(x)$ / $\forall x \text{ Lahori}(x) \rightarrow \text{Pakistani}(x)$ / $\text{Lahori}(\text{Ali})$ / $\text{Pakistani}(\text{Sara})$ / $\text{NOT Lahori}(\text{Sara})$

(a) Is Ali Pakistani? (b) Does Ali speak Urdu? (c) Does Sara speak Urdu? (d) Is Sara Lahori?

Q15. Write FOL sentences for a BANKING SYSTEM. Define predicates first:

(a) Every account holder must have a valid ID. (b) Some transactions are flagged. (c) No minor can open a savings account without a guardian. (d) All flagged transactions notify the account owner.

Q16. CHALLENGE KB: $\forall x \text{ Employee}(x) \text{ AND } \text{YearsExp}(x) \geq 5 \rightarrow \text{Senior}(x)$ / $\forall x \text{ Senior}(x) \text{ AND } \text{HasProject}(x) \rightarrow \text{GetsBonus}(x)$ / $\forall x \text{ GetsBonus}(x) \rightarrow \text{TaxDeducted}(x)$ / $\text{Employee}(\text{Zara}) \text{ YearsExp}(\text{Zara}) = 7$ $\text{HasProject}(\text{Zara})$ / $\text{Employee}(\text{Bilal}) \text{ YearsExp}(\text{Bilal}) = 3$ $\text{HasProject}(\text{Bilal})$

(a) Does Zara get a bonus? (b) Is Zara's tax deducted? (c) Does Bilal get a bonus? Why? (d) Write a new rule so Bilal also gets a benefit.

Practice Questions — Part 4: Answers

Q13. FOL translations (Birds)

(a) $\forall x: \text{Bird}(x) \rightarrow \text{CanFly}(x)$ (b) $\exists x: \text{Bird}(x) \text{ AND NOT } \text{CanFly}(x)$ (c) $\forall x: \text{Penguin}(x) \rightarrow \text{Bird}(x) \text{ AND NOT } \text{CanFly}(x)$ (d) $\forall x: \text{Animal}(x) \text{ AND NOT } \text{CanFly}(x) \rightarrow \exists p: \text{LivesAt}(x,p)$

Q14. KB inference — Pakistani / Lahori domain

(a) $\text{Lahori}(\text{Ali})=T + \text{Rule}(\text{Lahori} \rightarrow \text{Pakistani}) \Rightarrow \text{Pakistani}(\text{Ali})=T$ YES
(b) $\text{Pakistani}(\text{Ali})=T + \text{Rule}(\text{Pakistani} \rightarrow \text{SpeaksUrdu}) \Rightarrow \text{SpeaksUrdu}(\text{Ali})=T$ YES
(c) $\text{Pakistani}(\text{Sara})=T + \text{Rule} \Rightarrow \text{SpeaksUrdu}(\text{Sara})=T$ YES
(d) NOT Lahori(Sara) is stated directly in the KB $\Rightarrow$ NO

Q15. Banking System FOL

(a) $\forall x: \text{AccountHolder}(x) \rightarrow \text{HasValidID}(x)$ (b) $\exists t: \text{Transaction}(t) \text{ AND } \text{Flagged}(t)$ (c) $\forall x: \text{Minor}(x) \rightarrow (\text{CanOpenSavings}(x) \rightarrow \exists g: \text{Guardian}(x,g))$ (d) $\forall t,x: \text{Transaction}(t) \text{ AND } \text{Flagged}(t) \text{ AND } \text{Owner}(x,t) \rightarrow \text{Notifies}(t,x)$

Q16. CHALLENGE — Employee KB (Zara & Bilal)

(a) $\text{YearsExp}(\text{Zara})=7 \geq 5 \Rightarrow \text{Senior}(\text{Zara})=T. \text{Senior} + \text{HasProject} \Rightarrow \text{GetsBonus}(\text{Zara})=T$ YES
(b) $\text{GetsBonus}(\text{Zara})=T \Rightarrow \text{TaxDeducted}(\text{Zara})=T$ YES
(c) $\text{YearsExp}(\text{Bilal})=3 < 5 \Rightarrow \text{Senior}=\text{FALSE} \Rightarrow \text{Rule cannot fire} \Rightarrow \text{NO BONUS}$
(d) New rule: $\forall x: \text{Employee}(x) \text{ AND } \text{HasProject}(x) \rightarrow \text{GetsProjectAllowance}(x) \Rightarrow \text{Bilal qualifies!}$

05

The Wumpus World

Putting It All Together — Knowledge + Reasoning in Action

The most famous teaching example in AI. All concepts from Parts 1–4 appear simultaneously in one coherent problem.

5.1 & 5.2 — The Wumpus World Setup & Percepts

Introduced by Gregory Yob (1973), adopted by Russell & Norvig. A 4x4 grid cave with hidden dangers.

The Setup — 4x4 Grid Cave

 WUMPUS	A terrible monster. Walk into its room = you die.
 PITS	Bottomless pits. Fall in = you die.
★ GOLD	The treasure you are trying to collect.
▶ AGENT	You! Start at (1,1) bottom-left, facing right.
BLIND	You CANNOT see neighbouring rooms. Only sense current room.
GOAL	Find the gold, pick it up, return to (1,1) alive.

5.2 Percepts — What the Agent Can Sense

STENCH	You smell something awful.	→ Wumpus is ADJACENT to you (but not here).
BREEZE	You feel a draft.	→ A Pit is ADJACENT to you.
GLITTER	Something shines.	→ Gold is in THIS EXACT room — grab it!
BUMP	You hit a wall.	→ You tried to step outside the grid.
SCREAM	A terrible shriek.	→ Your arrow killed the Wumpus.

5.3 World Rules (Fixed Knowledge Base)

W1: $\text{Stench}(x,y) \leftrightarrow \text{Wumpus adjacent to } (x,y)$ **W2:** $\text{Breeze}(x,y) \leftrightarrow \text{Pit adjacent to } (x,y)$ **W3:** $\text{Glitter}(x,y) \leftrightarrow \text{Gold}(x,y)$ **W4:** Agent dies if at (x,y) AND $[\text{Wumpus}(x,y) \text{ OR } \text{Pit}(x,y)]$ **W5:** $\text{OK}(x,y) \leftrightarrow \text{NOT Pit}(x,y) \text{ AND NOT Wumpus}(x,y)$

5.4 The Standard Wumpus World Map

Standard configuration: Wumpus at (1,4) | Pit at (3,1), (3,3), (3,4) | Gold at (2,4). Agent starts at (1,1).

(1,4) Wumpus 🦇 [Stench nearby]	(2,4) GOLD ★ [Stench + Breeze]	(3,4) PIT ⚠️ [Breeze nearby]	(4,4) [Breeze]	Row 4
(1,3) [Stench]	(2,3) [Breeze]	(3,3) PIT ⚠️ [Breeze nearby]	(4,3) [Breeze]	Row 3
(1,2) [Stench]	(2,2) [Nothing]	(3,2) [Breeze]	(4,2) [Nothing]	Row 2
(1,1) AGENT ▶ START [Nothing]	(2,1) [Breeze]	(3,1) PIT ⚠️ [Breeze nearby]	(4,1) [Nothing]	Row 1

5.5 The Agent Reasons — Step by Step

Agent starts at (1,1) with empty KB. Sense → Update KB → Infer → Decide → Move.

Step 1 — Agent at (1,1)

Percept: [Nothing — No Stench, No Breeze, No Glitter]

Decision: Move to (2,1) — confirmed safe.

Rule W2: NOT Breeze(1,1) → NOT Pit(1,2) AND NOT Pit(2,1)
Rule W1: NOT Stench(1,1) → NOT Wumpus(1,2) AND NOT Wumpus(2,1)

KB Update: OK(1,2)=TRUE,
OK(2,1)=TRUE

Step 2 — Agent at (2,1)

Percept: [No Stench, BREEZE, No Glitter]

Decision: Go BACK to (1,1). Don't risk (3,1) or (2,2) yet.

Rule W2: Breeze(2,1)=TRUE → Pit adjacent to (2,1)
Adjacent: (1,1), (3,1), (2,2). We know OK(1,1)=TRUE.
Therefore: Pit(3,1) OR Pit(2,2) — uncertain!

KB Update:
PitPossible(3,1)=TRUE,
PitPossible(2,2)=TRUE

Step 3 — Agent at (1,2)

Percept: [STENCH, No Breeze, No Glitter]

Decision: Move to (2,2) — confirmed safe!
Agent never SAW Pit or Wumpus. It DEDUCED their locations.

Rule W1: Stench(1,2)=TRUE → Wumpus adjacent. Options: (2,2) or (1,3) or (1,1)X → Wumpus(2,2) OR Wumpus(1,3).
Rule W2: NOT Breeze(1,2) → NOT Pit(2,2).
COMBINE: Pit(3,1) OR Pit(2,2) AND NOT Pit(2,2) ⇒ Pit(3,1)=CONFIRMED!
NOT Stench(2,1) → NOT Wumpus(2,2). From (1,2): Wumpus in (1,3). From (1,3): Stench→Wumpus in (1,4) or (2,3). (2,3) has no stench → Wumpus(1,4)=CONFIRMED!

KB Update:
Pit(3,1)=CONFIRMED,
Wumpus tentative at (1,3)/(1,4), OK(2,2)=TRUE

Complete Solution — Steps 4 & 5: Safe at (2,2), Shooting Wumpus

KB so far: $\text{Pit}(3,1)=\checkmark$ Wumpus tentative(1,4) | Each step: Percept $\rightarrow$ Prop. Sentence $\rightarrow$ Modus Ponens $\rightarrow$ FOL equivalent $\rightarrow$ Decision

Step 4

Room
(2,2)

Percept: NOTHING — No
Stench, No Breeze, No Glitter

$\rightarrow$ OK(2,3) confirmed safe.
But to reach (2,4) we need to
pass (2,3). Must also confirm
Wumpus(1,4) before
approaching (2,4).

Prop. Sentences:

$\text{NOT } S[2][2]=\text{TRUE} \Leftrightarrow \text{NOT } (W[1][2] \text{ OR } W[3][2] \text{ OR } W[2][1] \text{ OR } W[2][3]) \rightarrow$ No Wumpus adjacent

$\text{NOT } B[2][2]=\text{TRUE} \Leftrightarrow \text{NOT } (P[1][2] \text{ OR } P[3][2] \text{ OR } P[2][1] \text{ OR } P[2][3]) \rightarrow$ No Pit adjacent

Modus Ponens:

$\text{NOT } S[2][2]=\text{TRUE} + \text{Rule W1} \rightarrow$ NOT Wumpus adjacent to (2,2) $\checkmark$ [Wumpus at (1,4) confirmed —
NOT adjacent to (2,2)]

$\text{NOT } B[2][2]=\text{TRUE} + \text{Rule W2} \rightarrow$ NOT Pit in ANY adjacent room $\rightarrow$ NOT Pit(1,2) $\checkmark$ NOT Pit(3,2) $\checkmark$ NOT
Pit(2,1) $\checkmark$ NOT Pit(2,3) $\checkmark$

Therefore: OK(1,2)=TRUE, OK(3,2)=TRUE, OK(2,1)=TRUE, OK(2,3)=TRUE — ALL four neighbours
confirmed safe!

FOL:

$\forall x \forall y: \text{Stench}(x,y) \Leftrightarrow \exists a \exists b: \text{Adjacent}(x,y,a,b) \wedge$
Wumpus(a,b)

Instantiate at (2,2): NOT Stench(2,2) $\rightarrow \neg \exists$ Wumpus
adjacent $\checkmark$

$\forall x \forall y: \text{Breeze}(x,y) \Leftrightarrow \exists a \exists b: \text{Adjacent}(x,y,a,b) \wedge \text{Pit}(a,b)$
Instantiate at (2,2): NOT Breeze(2,2) $\rightarrow \neg \exists$ Pit adjacent
 $\rightarrow$ OK(2,3)=CONFIRMED SAFE $\checkmark$

KB State:

OK(2,2)=TRUE $\checkmark$
OK(2,3)=TRUE $\checkmark$
OK(3,2)=TRUE $\checkmark$
Wumpus(1,4) tentative
Gold at (2,4)

Step 5

Room
SHOOT ARROW $\rightarrow$
(1,4)

Percept: ACTION: Fire arrow
toward (1,4). SCREAM heard!

$\rightarrow$ Move (2,2) $\rightarrow$ (2,3) $\rightarrow$ (2,4).
Wumpus dead, OK(2,3)
confirmed, Gold at (2,4).

Prop. Sentences:

Scream=TRUE $\rightarrow$ WumpusDead=TRUE

WumpusDead=TRUE $\rightarrow$ Wumpus(1,4) eliminated — no more death threat from Wumpus

Modus Ponens:

Scream=TRUE + Rule W [Scream $\rightarrow$ WumpusDead] $\checkmark$

WumpusDead=TRUE $\rightarrow$ Wumpus(1,4) dead; (1,4) still has stench residue but no live threat
NOT Breeze(2,2)=TRUE $\rightarrow$ NOT Pit(2,3)=TRUE $\checkmark$ [established in Step 4]

Path to Gold: (2,2) $\rightarrow$ (2,3) $\rightarrow$ (2,4). OK(2,3)=TRUE $\checkmark$. OK(2,4)? Breeze at (2,3) means Pit adj to (2,3) —
Pit(3,3) [confirmed from map]. NOT Pit(2,4)=safe to enter.

Wumpus(1,4)=DEAD $\checkmark$

FOL:

$\forall x \forall y: \text{Scream} \rightarrow \neg \exists a \exists b: \text{LiveWumpus}(a,b)$ [after
arrow fired at (1,4)]

Instantiate: Scream=TRUE $\rightarrow$ WumpusDead=TRUE

W5 FOL: OK(2,3) $\Leftrightarrow \neg \text{Pit}(2,3) \wedge \neg \text{Wumpus}(2,3)$

$\neg \text{Pit}(2,3)=\text{TRUE}$ [NOT Breeze(2,2)];

$\neg \text{Wumpus}(2,3)=\text{TRUE}$ [Wumpus at (1,4), now dead] $\rightarrow$
OK(2,3)=TRUE $\checkmark$

KB State:

WumpusDead=TRUE
OK(2,3)=TRUE $\checkmark$
OK(2,4) assessed: $\neg \text{Wumpus}(2,4)$ [dead]
Gold at (2,4) — advance!

Steps 6-7: Via (2,3) to (2,4) and Grabbing the Gold

The agent reaches the gold room — propositional proof, Modus Ponens chain, and FOL inference all shown.

Step 6 — Agent at (2,3) → Breeze=TRUE [Pit(3,3) adj]. Then (2,4): Stench=TRUE [Wumpus adj,dead], Breeze=TRUE [Pit(3,4)], GLITTER=TRUE

Propositional Sentences:

Glitter[2][3] = TRUE $\leftrightarrow$ Gold[2][3] = TRUE [World Rule W3]

Breeze[2][3] = TRUE $\rightarrow$ Pit adjacent to (2,3): Pit(3,3)=confirmed

Glitter[2][4] = TRUE $\leftrightarrow$ Gold[2][4] = TRUE [World Rule W3]

Stench[2][4]=TRUE $\rightarrow$ Wumpus adjacent (1,4)=DEAD $\checkmark$, Breeze[2][4]=TRUE $\rightarrow$ Pit(3,4) confirmed

(* Pit(3,4) confirmed from map; ** (2,4) IS the gold room = this room)

Modus Ponens Chain:

Glitter(2,4)=TRUE + Rule W3 [Glitter(x,y) $\leftrightarrow$ Gold(x,y)]

$\Rightarrow$ Gold(2,4) = TRUE [Modus Ponens] ★ GRAB IT!

Breeze(2,3)=TRUE + Rule W2 $\rightarrow$ Pit adj to (2,3) $\rightarrow$ Pit(3,3)=CONFIRMED

$\Rightarrow$ Pit(3,3)=CONFIRMED [only unconfirmed adjacent pit]

First Order Logic — Generalised Rules Applied at (2,3)

W3 FOL: $\forall x \forall y: \text{Glitter}(x,y) \leftrightarrow \text{Gold}(x,y) \rightarrow$ Instantiate $x=2,y=4: \text{Glitter}(2,4)=\text{TRUE} \Rightarrow \text{Gold}(2,4)=\text{TRUE} \checkmark$

W2 FOL: $\forall x \forall y: \text{Breeze}(x,y) \leftrightarrow \exists a \exists b: \text{Adjacent}(x,y,a,b) \wedge \text{Pit}(a,b) \rightarrow$ Instantiate $x=2,y=3: \text{Breeze}(2,3)=\text{TRUE} \Rightarrow \exists \text{ Pit adjacent} \rightarrow \text{Pit}(3,3)=\text{CONFIRMED} \checkmark$

W2 FOL: $\forall x \forall y: \text{Glitter}(x,y) \leftrightarrow \text{Gold}(x,y) \rightarrow$ Instantiate $x=2,y=4: \text{Glitter}(2,4)=\text{TRUE} \Rightarrow \text{Gold}(2,4)=\text{TRUE} \checkmark$ GRAB IT!

★ GOLD GRABBED! Agent now holds the Gold at (2,4) ★

KB State: Gold=GRABBED | OK(2,4)=TRUE | Wumpus(1,4)=DEAD | Pit(3,1)=CONFIRMED | Pit(3,3)=CONFIRMED | Pit(3,4)=CONFIRMED

Complete Solution — Step 7: Safe Return Path to (1,1)

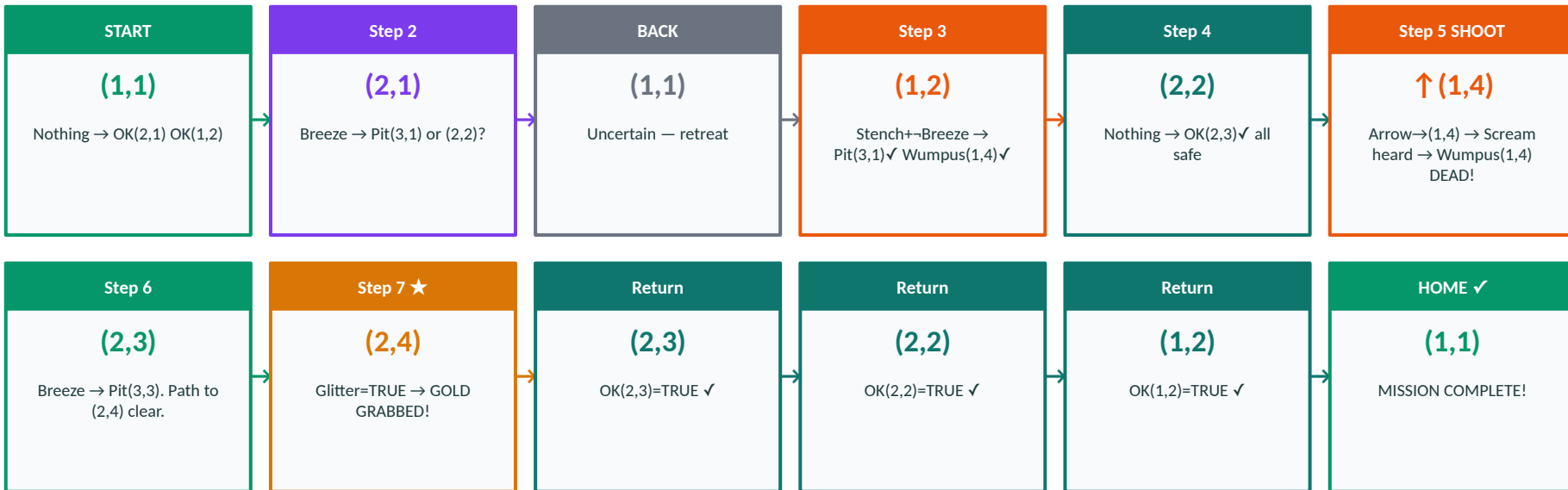
Agent has Gold. Confirmed safe path: (2,4)→(2,3)→(2,2)→(1,2)→(1,1). Each step verified by OK() reasoning.

KB State	Confirmed SAFE: (1,1) (2,1) (1,2) (2,2) (2,3) (2,4) Confirmed DEADLY: Pit(3,1) Pit(3,3) Pit(3,4) Wumpus(1,4)[dead] Gold: GRABBED ★	
(2,4) ↓ (2,3)	Prop: OK[2][3] = TRUE [NOT Breeze(2,2)=TRUE → NOT Pit adjacent to (2,2) → NOT Pit(2,3)] NOT Pit(2,3)=TRUE AND NOT Wumpus(2,3)=TRUE [Wumpus at (1,4), not (2,3)]	MP: NOT Breeze(2,2)=TRUE + Rule W2 → ¬∃Pit adj(2,2) → NOT Pit(2,3)=TRUE Wumpus(1,4)=DEAD, not at (2,3) → NOT Wumpus(2,3)=TRUE → OK(2,3)=TRUE ✓ FOL: W2 FOL at (2,2): ¬Breeze(2,2) → ¬∃a∃b: Adjacent(2,2,a,b) ∧ Pit(a,b) → ¬Pit(2,3) ✓ W5 FOL: ¬Pit(2,3) ∧ ¬Wumpus(2,3) → OK(2,3)=TRUE ✓
(2,3) ↓ (2,2)	Prop: OK[2][2] = TRUE [percept at (2,2) = Nothing → NOT Pit AND NOT Wumpus adjacent] NOT Pit(2,2)=TRUE AND NOT Wumpus(2,2)=TRUE	MP: NOT Breeze(2,2)=TRUE + Rule W2 → NOT Pit adj to (2,2) → NOT Pit(2,2)=TRUE ✓ NOT Stench(2,2)=TRUE → NOT Wumpus adj → NOT Wumpus(2,2)=TRUE → OK(2,2)=TRUE ✓ FOL: W1+W2 FOL at (2,2): ¬Stench(2,2) ∧ ¬Breeze(2,2) → ¬∃Wumpus adj ∧ ¬∃Pit adj ✓ W5 FOL: ¬Pit(2,2) ∧ ¬Wumpus(2,2) → OK(2,2)=TRUE ✓
(2,2) ↓ (1,2)	Prop: OK[1][2] = TRUE [NOT Breeze(1,2) → NOT Pit adjacent. Stench(1,2)=TRUE but Wumpus at (1,4) not (1,2)] NOT Pit(1,2)=TRUE AND NOT Wumpus(1,2)=TRUE	MP: NOT Breeze(1,2)=TRUE + Rule W2 → NOT Pit adj to (1,2) → NOT Pit(1,2)=TRUE ✓ Stench(1,2)=TRUE; Wumpus confirmed at (1,4) not (1,2) → NOT Wumpus(1,2)=TRUE → OK(1,2)=TRUE ✓ FOL: W2 FOL at (1,2): ¬Breeze(1,2) → ¬∃Pit adj(1,2) → ¬Pit(1,2) ✓ W5 FOL: ¬Pit(1,2) ∧ ¬Wumpus(1,2) [Wumpus at (1,4)] → OK(1,2)=TRUE ✓

	Prop: OK[1][1] = TRUE [starting room — agent was alive here at Step 1]	MP: Step 1: NOT Breeze(1,1) → NOT Pit(1,2) AND NOT Pit(2,1)
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Complete Solution — Mission Summary: Full Path with Logic at Every Step

Standard map: Wumpus(1,4) | Pit(3,1) | Pit(3,3) | Pit(3,4) | Gold(2,4). Agent starts at (1,1).



What Made This Possible — The Three Layers of Logic Used at Every Step

Propositional Logic

$B[2][1]=\text{TRUE}$, $S[1][2]=\text{TRUE}$, $OK[2][2]=\text{TRUE}$ — each room state stored as a T/F symbol. The language of the KB.

Modus Ponens

IF rule AND observed fact THEN conclusion. Applied at every single step to chain sensory input into safe decisions.

First Order Logic

$\forall x \forall y + \exists a \exists b$: ONE universal rule replaces 16 propositional rules. Scales to any grid. Enables proof by Universal Instantiation.

5.5 Core Insight + 5.6 Propositional Logic Encoding

The agent stays alive because it REASONS before it acts — not because it was lucky.

THE CORE INSIGHT — Reasoning Without Direct Observation

At Step 3, the agent NEVER SAW the Pit at (3,1) or the Wumpus at (1,4). It DEDUCED both locations by combining observations across multiple rooms.

- Store what you observe as T/F propositions in your KB
- Apply fixed world rules as logical sentences
- Use Modus Ponens + Resolution to derive new facts
- The agent stays alive because it REASONS before it acts

5.6 Propositional Symbols

$B[c][r]$ = Breeze at column c , row r

$S[c][r]$ = Stench at column c , row r

$P[c][r]$ = Pit at column c , row r

$W[c][r]$ = Wumpus at column c , row r

$OK[c][r]$ = Safe to visit column c , row r

World Rules as Propositions

$B[1][1] \leftrightarrow (P[1][2] \vee P[2][1])$

$B[2][1] \leftrightarrow (P[1][1] \vee P[3][1] \vee P[2][2])$

$B[1][2] \leftrightarrow (P[1][1] \vee P[2][2] \vee P[1][3])$

$S[1][2] \leftrightarrow (W[1][1] \vee W[2][2] \vee W[1][3])$

$OK[c][r] \leftrightarrow \neg P[c][r] \wedge \neg W[c][r]$

Observed Facts:

$\neg B[1][1]=T, \neg S[1][1]=T \mid B[2][1]=T$

$S[1][2]=T, \neg B[1][2]=T$

5.7a — From Propositional Logic to First Order Logic in Wumpus World

Why propositional logic alone is not enough — and how FOL solves it.

The Problem with Propositional Logic for Wumpus World

In a 4x4 grid we wrote ONE rule per room:

$B[1][1] \leftrightarrow (P[1][2] \vee P[2][1])$ $B[2][1] \leftrightarrow (P[1][1] \vee P[3][1] \vee P[2][2])$ $B[1][2] \leftrightarrow \dots$ (16 separate rules, one per cell!)

What if the grid were 100x100? We would need 10,000 separate propositional rules. And the rules for a 4x4 grid cannot be reused for ANY other grid size.

The FOL Solution — ONE Rule Covers ALL Rooms in ANY Grid

Instead of writing $B[1][1] \leftrightarrow \dots$, $B[2][1] \leftrightarrow \dots$, $B[1][2] \leftrightarrow \dots$ separately, FOL lets us write:

$\forall x \forall y: \text{Breeze}(x,y) \leftrightarrow \exists a \exists b: \text{Adjacent}(x,y,a,b) \wedge \text{Pit}(a,b)$

Meaning: For ALL rooms (x,y): there is a Breeze at (x,y) IF AND ONLY IF there EXISTS some adjacent room (a,b) that contains a Pit.

This ONE sentence replaces ALL 16 propositional rules and works for any grid size.

$\forall$ Universal Quantifier

Speaks about ALL rooms at once — not just (1,1), (2,1), (1,2) etc. one by one.

$\exists$ Existential Quantifier

Says there EXISTS some adjacent room with a Pit — we don't need to name each room.

⚡ Variables x,y,a,b

Placeholders for any room coordinates — the rule generalises to any grid, any map.

🔗 Predicates & Relations

$\text{Adjacent}(x,y,a,b)$ is a relationship — something impossible to express in propositional logic.

5.7b — All Wumpus World Rules in First Order Logic

The complete FOL knowledge base — one sentence per rule, valid for any grid size.

W1 — Stench Rule

Prop: $S[x][y] \leftrightarrow (W[x-1][y] \vee W[x+1][y] \vee W[x][y-1] \vee W[x][y+1])$

FOL: $\forall x \forall y: \text{Stench}(x,y) \leftrightarrow \exists a \exists b: \text{Adjacent}(x,y,a,b) \wedge \text{Wumpus}(a,b)$

There is a stench at (x,y) if and only if the Wumpus is in SOME adjacent room (a,b). The $\exists$ quantifier handles all 4 neighbours at once without naming them.

W2 — Breeze Rule

Prop: $B[x][y] \leftrightarrow (P[x-1][y] \vee P[x+1][y] \vee P[x][y-1] \vee P[x][y+1])$

FOL: $\forall x \forall y: \text{Breeze}(x,y) \leftrightarrow \exists a \exists b: \text{Adjacent}(x,y,a,b) \wedge \text{Pit}(a,b)$

There is a breeze at (x,y) if and only if SOME adjacent room (a,b) contains a Pit. Same pattern as W1 — one formula covers all grid positions.

W3 — Glitter Rule

Prop: $\text{Glitter}[x][y] \leftrightarrow \text{Gold}[x][y]$

FOL: $\forall x \forall y: \text{Glitter}(x,y) \leftrightarrow \text{Gold}(x,y)$

Gold glitters exactly where it sits. Both propositional and FOL versions are similar — but FOL version scales to any room without listing each cell.

W4 — Death Rule

Prop: $\text{Dies} \leftrightarrow (\text{Agent at same room as Wumpus OR Pit})$

FOL: $\forall x \forall y: \text{Agent}(x,y) \wedge (\text{Wumpus}(x,y) \vee \text{Pit}(x,y)) \rightarrow \text{Dies}$

If the agent is at (x,y) AND either the Wumpus OR a Pit is also at (x,y), then the agent dies. The $\forall$ universally covers every possible room location.

W5 — Safety Rule

Prop: $\text{OK}[x][y] \leftrightarrow \text{NOT } P[x][y] \text{ AND NOT } W[x][y]$

FOL: $\forall x \forall y: \text{OK}(x,y) \leftrightarrow \neg \text{Pit}(x,y) \wedge \neg \text{Wumpus}(x,y)$

Room (x,y) is safe to enter if and only if it contains neither a Pit nor the Wumpus. This rule is used at every move decision.

5.7c — FOL Inference in Action: Proving Pit(3,1)

Applying Universal Instantiation + Modus Ponens using the FOL encoding of Wumpus World.

Step 1 — Start with the Universal FOL Rule (W2)

FOL Rule W2: $\forall x \forall y: \text{Breeze}(x,y) \leftrightarrow \exists a \exists b: \text{Adjacent}(x,y,a,b) \wedge \text{Pit}(a,b)$

Translation: For EVERY room (x,y) in the grid — if there is a Breeze there, then SOME adjacent room has a Pit, and vice versa.

Step 2 — Universal Instantiation: substitute x=2, y=1

Substitute x=2, y=1 into the universal rule:

Breeze(2,1) $\leftrightarrow \exists a \exists b: \text{Adjacent}(2,1,a,b) \wedge \text{Pit}(a,b)$

The Adjacent predicate for room (2,1) holds for: (1,1), (3,1), (2,2) — these are its three neighbours.

Step 3 — Apply Modus Ponens with observed fact

Observed fact: Breeze(2,1) = TRUE [agent sensed breeze at room (2,1)]

Apply Modus Ponens: Breeze(2,1)=TRUE + Rule W2 instantiated at (2,1) $\Rightarrow$

$\exists a \exists b: \text{Adjacent}(2,1,a,b) \wedge \text{Pit}(a,b) = \text{TRUE}$

Conclusion: Pit(1,1) OR Pit(3,1) OR Pit(2,2) — at least one of the three adjacent rooms contains a Pit.

Step 4 — Combine with evidence from room (1,2): Resolution $\Rightarrow$ Pit(3,1) CONFIRMED

Instantiate W2 at (1,2): NOT Breeze(1,2) = TRUE $\Rightarrow$ NOT (Pit(1,1) OR Pit(2,2) OR Pit(1,3)) $\Rightarrow$ NOT Pit(2,2) = TRUE

From Step 3: Pit(1,1) OR Pit(3,1) OR Pit(2,2). NOT Pit(1,1) [came from (1,1), safe]. NOT Pit(2,2) [from above].

Resolution: The only remaining option is **Pit(3,1) = TRUE ✓ CONFIRMED!**

5.7 — Why Wumpus World Ties the Whole Course Together

DIKW Pyramid	Percept data → room info → pit/wumpus locations (knowledge) → safe path home (wisdom)
Declarative Knowledge	Rules W1-W5 are stored facts in the KB — pure declarative knowledge
Procedural Knowledge	The agent's exploration strategy is a procedure: sense → infer → act
Propositional Logic	Every percept and room status is a T/F proposition ($B[2][1]=\text{TRUE}$, etc.)
Knowledge Base	The agent builds and updates its KB as it explores each room
Modus Ponens	Used at every step: IF world rule AND percept THEN derive conclusion
Entailment	The agent asks: does my KB entail that room X is safe to enter?
Resolution	Used in Step 3 to eliminate Pit possibilities across multiple rooms
FOL Extension	$\forall x \forall y: \text{Breeze}(x,y) \leftrightarrow \exists a \exists b: \text{Adjacent}(x,y,a,b) \wedge \text{Pit}(a,b)$ — ONE rule replaces all 16 propositional breeze rules. See slides 5.7a–5.7c for the full FOL treatment.

Class Activity 8 — Wumpus World Reasoning Challenge

Standard map: Pit(3,1), Pit(3,3), Pit(3,4), Wumpus(1,4), Gold(2,4). Group of 4 students.

Tasks

TASK 1: List what percepts the agent receives at each room: (1,1), (2,1), (1,2), (2,2)

TASK 2: After visiting (1,1) → (2,1) → (1,2) in that order:

- a) What can the agent conclude about Pit locations?
- b) What can the agent conclude about the Wumpus location?
- c) Which rooms are confirmed SAFE? Which are DANGEROUS?
- d) Is it safe to move to (2,2)? Write the propositional logic justification.

TASK 3: Agent enters (2,2). Percept = [Stench=FALSE, Breeze=FALSE, Glitter=FALSE, Bump=FALSE, Scream=FALSE]

- e) What new information does this add to the KB?
- f) Is the Wumpus location now confirmed? How?
- g) What is the safest path from (2,2) to the Gold at (2,4)? Justify by checking OK() for each room.

BONUS: Write 3 propositional logic sentences that together prove NOT Pit(2,2) is TRUE.

Class Activity 8 — Solution

Wumpus World Reasoning Challenge — Teacher Guide

Tasks 1 & 2

Task 1 Percepts:

(1,1): Nothing | (2,1): Breeze | (1,2): Stench | (2,2): NOTHING

a) Pit:

$B[2][1]=T \Rightarrow$ Pit in $\{(3,1),(2,2)\}$. $\text{NOT } B[1][2]=T \Rightarrow \text{NOT Pit}(2,2) \Rightarrow \text{Pit}(3,1) = \text{CONFIRMED}$

b) Wumpus:

$S[1][2]=T \Rightarrow$ Wumpus in $\{(2,2),(1,3)\}$. $\text{NOT } S[2][1]=T \Rightarrow \text{NOT Wumpus}(2,2) \Rightarrow$ Wumpus tentative at (1,3)/(1,4). More steps needed $\rightarrow$ Wumpus(1,4) CONFIRMED via (1,3) Stench.

c) Safe: (1,1),(2,1),(1,2),(2,2). Dangerous: (3,1)=PIT (1,4)=WUMPUS (3,3)=PIT (3,4)=PIT

d) $\text{NOT Pit}(2,2)=\text{TRUE}$ AND $\text{NOT Wumpus}(2,2)=\text{TRUE} \Rightarrow \text{OK}(2,2)=\text{TRUE}$ by Rule W5. **SAFE TO ENTER.**

Task 3 & Bonus

e) Nothing at (2,2):

$\text{NOT Stench}(2,2) \rightarrow$ Wumpus NOT adjacent to (2,2) ✓ [Wumpus at (1,4) is not adjacent]

$\text{NOT Breeze}(2,2) \rightarrow$ NO Pit in ANY adjacent room $\rightarrow \text{NOT Pit}(2,3)=\text{TRUE}$, $\text{NOT Pit}(3,2)=\text{TRUE}$ ✓

$\rightarrow \text{OK}(2,3)=\text{TRUE}$, $\text{OK}(3,2)=\text{TRUE}$, $\text{OK}(2,1)=\text{TRUE}$, $\text{OK}(1,2)=\text{TRUE}$ — ALL neighbours safe!

f) Wumpus(1,4) = CONFIRMED:

$\text{Stench}(1,2)=T \rightarrow$ Wumpus in $\{(2,2),(1,3)\}$. $\text{NOT Stench}(2,1) \rightarrow \text{NOT Wumpus}(2,2)$.

Visit (1,3): $\text{Stench}(1,3)=T \rightarrow$ Wumpus in (1,4) or (2,3). $\text{NOT Stench}(2,3) \rightarrow$ Wumpus(1,4) = CONFIRMED ✓

g) Safest path (2,2) $\rightarrow$ Gold(2,4):

$\text{OK}(2,3)=\text{TRUE}$ [NOT Breeze(2,2)]. Shoot arrow at (1,4) $\rightarrow$ Scream $\rightarrow$ Wumpus DEAD.

Move (2,2) $\rightarrow$ (2,3) $\rightarrow$ (2,4) grab Gold. Return: (2,4) $\rightarrow$ (2,3) $\rightarrow$ (2,2) $\rightarrow$ (1,2) $\rightarrow$ (1,1). DONE!

BONUS — 3 sentences proving NOT Pit(2,2):

1. $B[1][2] \leftrightarrow (P[1][1] \text{ OR } P[2][2] \text{ OR } P[1][3])$ [World Rule W2]

2. $\text{NOT } B[1][2] = \text{TRUE}$ [observed: no breeze at (1,2)]

3. Contrapositive: $\text{NOT } B[1][2] \rightarrow \text{NOT } P[1][1] \text{ AND NOT } P[2][2] \text{ AND NOT } P[1][3]$

Apply Modus Ponens on 2+3 $\Rightarrow$ NOT Pit(2,2)=TRUE. PROVED. ✓

Practice Questions — Part 5: Wumpus World

Think carefully and write your answers before turning to the next slide.

Q17. The agent is at (1,1) and detects a Breeze. Write the full propositional sentence for this percept, apply Modus Ponens, and state exactly what the agent should NOT do and why.

Q18. The agent is at (2,2). The percept is [Nothing — Stench=FALSE, Breeze=FALSE]. List all 4 adjacent rooms, and for each room state whether it is SAFE, RISKY, or UNKNOWN — with full propositional justification.

Q19. Why is the Wumpus World called a 'knowledge-based agent' problem? In what specific ways is it different from a robot that simply follows a fixed set of programmed instructions?

Q20. CHALLENGE: The agent has reached (2,4) and found the Gold. The following facts are confirmed in the KB: Wumpus(1,4)=TRUE[dead], Pit(3,1)=TRUE, Pit(3,3)=TRUE, Pit(3,4)=TRUE.

Plan the complete safe return path back to (1,1). Justify every single step by checking the OK() condition for each room visited.

Practice Questions — Part 5: Answers

Q17. Breeze at (1,1) — propositional reasoning

Rule: $B[1][1] \Leftrightarrow (P[1][2] \vee P[2][1])$. Fact: $B[1][1]=\text{TRUE}$. Apply Modus Ponens: $\text{TRUE} \Rightarrow P[1][2] \vee P[2][1]=\text{TRUE}$ [at least one is a Pit].
Do NOT move to (1,2) or (2,1) until more evidence eliminates one. Safest: stay put and seek info via another route.

Q18. Agent at (2,2), Percept=Nothing [Stench=F, Breeze=F]

Adjacent: (1,2), (3,2), (2,1), (2,3).

NOT Breeze(2,2)=TRUE + Rule W2 $\rightarrow$ NO Pit adjacent $\rightarrow$ NOT Pit(1,2) ✓ NOT Pit(3,2) ✓ NOT Pit(2,1) ✓ NOT Pit(2,3) ✓

NOT Stench(2,2)=TRUE + Rule W1 $\rightarrow$ NO Wumpus adjacent $\rightarrow$ NOT Wumpus in any of them.

$\rightarrow$ ALL FOUR rooms: pit-free AND wumpus-free $\rightarrow$ OK(1,2)=SAFE ✓ OK(3,2)=SAFE ✓ OK(2,1)=SAFE ✓ OK(2,3)=SAFE ✓

Q19. KB agent vs fixed-instruction robot

KB Agent: BUILDS and UPDATES a world model. Collects percepts as logical facts. Applies inference rules to derive facts never directly observed. Makes rational decisions based on accumulated knowledge.

Fixed-instruction robot: follows pre-programmed responses. Cannot adapt — if never told 'pit at (3,1)', cannot know. Cannot reason from combinations of observations.

The KRR difference: the Wumpus agent KNOWS (pit at 3,1 confirmed by logic) and plans accordingly. A reactive robot cannot do this.

Q20. CHALLENGE — Safe return path from (2,3) to (1,1)

Confirmed safe rooms: (1,1),(2,1),(1,2),(2,2),(2,3) [arrived here alive].

Return path: (2,3) $\rightarrow$ (2,2) $\rightarrow$ (1,2) $\rightarrow$ (1,1)

- (2,4) $\rightarrow$ (2,3): OK(2,3)=TRUE [NOT Breeze at (2,2) $\rightarrow$ no pit adj to (2,2) $\rightarrow$ NOT Pit(2,3)]
- (2,2) $\rightarrow$ (1,2): OK(1,2)=TRUE [no pit from Step 3; Wumpus confirmed at (1,4) not (1,2)]
- (1,2) $\rightarrow$ (1,1): OK(1,1)=TRUE [starting room, always safe]

Lecture Summary — Knowledge Representation & Reasoning

Course Overview	KRR bridges raw data and intelligent action. Powers Google, Siri, medical AI, robotics, Prolog. Foundation of explainable and neuro-symbolic AI.
Part 1: Intro to KR	DIKW: Data→Info→Knowledge→Wisdom. Real World→KB→Reasoning→Action. Representation Hypothesis: symbols + rules = intelligence.
Part 2: Types of Knowledge	Declarative (WHAT), Procedural (HOW), Heuristic (USUALLY WORKS), Meta (ABOUT KNOWLEDGE). Choosing the right type shapes the whole system design.
Part 3: Propositional Logic	T/F propositions. Five connectives ($\wedge \vee \neg \rightarrow \leftrightarrow$). Knowledge Bases. Modus Ponens inference. Limitation: cannot generalise over categories.
Part 4: First Order Logic	Variables, constants, predicates, functions. $\forall$ and $\exists$ quantifiers. Universal Instantiation + Modus Ponens. One rule covers all cases. Language of Prolog and ontologies.
Part 5: Wumpus World	All concepts combined. KB built step by step from percepts. Modus Ponens + Resolution derive hidden truths. Agent stays alive by reasoning, not luck.